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To make the parser recognize a 'while' loop, I would add a new production rule that defines its structure and connects it to existing statement rules. Specifically, I'd introduce a 'while_stmt' rule in the grammar that matches the keyword 'while', followed by a condition in parentheses and a statement or block to execute repeatedly. For example, the rule would look like:

```
while_stmt : WHILE '(' expr ')' stmt { $$ = createWhileStmt($3, $5); };
```

This rule tells the parser to create a 'WHILE' node in the Abstract Syntax Tree (AST) using the existing 'createWhileStmt' function, which stores both the condition and body of the loop. Then, I'd include 'while_stmt' in the main 'stmt' rule so it's treated like any other statement type. Since the AST and code generation files already define and handle 'NODE_WHILE', no major updates would be needed beyond this grammar addition. This allows the compiler to recognize, represent, and generate code for loops properly.